

SERVICE MANUAL



MEDITECH ABPM-05



Operation of Meditech ABPM-05

The device's intended use is to sequentially measure blood pressure and store the results over a period of 24 hours, during the normal daily activity of the patient.

ABPM-05 is a battery operated electronic device with a power source of 2 AA sized batteries or a NiMH accumulator.

The input of the DC-DC transformer with IC13 switch mode is connected to the power source via F2 fuse. It provides a VSTAB voltage of approximately 5V which supplies the IC8 supervisor microcontroller, the LCD1 LCD driver and the two analogous switches IC: IC9 and IC15. The same voltage charges the button cell B1 through D2 diode and R12 resistance. When there is no operating power source in the device, the B1 button cell supplies the IC power needs (VORA) of IC6 clock.

The digital part of the measurement electronics operate from the VKAP voltage, which is the same as the VSTAB voltage, but it can be switched off by the IC8 supervisor microcontroller through OFF sign to the T3 transistor. The electronics of pressure measurement operate through VANA power supply, which is created from a VKAP sign by the IC16 fix 4V stabilisator.

High current consumption components (motor pump and valves) operate from the VPOWER voltage. This voltage is produced by the IC14 DC-DC transformer and it is usually 3.6 V, but the IC1 main microcontroller can increase it up to 4.6 V via a GEAR sign. The DC-DC transformer is connected to the power source via the F1 fuse. It operates only, if the main microcontroller is switched on via VPEN sign and only if the supervisor microcontroller allows it through a VPWR sign. In case of switching it off, the operation of the motor and the valve is attached via the T2 FET as well. This dual cut off is a safety function, so that it will also happen in case of any component failure.

The supply (VRXD) of the T4,T5 transistors is normally between 4.5 and 5 V. There is an **off** mode of the device, when it's „totally“ switched off. In this case the supervisor microcontroller IC8 switches even this supply off in order to decrease the current consumption.

In the **stand-by** mode of the device, the supervisor microcontroller IC8 regularly reads the status of the two pushbuttons of the device, the TIMER interrupt line of the real-time clock IC6, and the presence of any signal on the optical serial line. If either pushbutton is pressed, or the real-time clock IC6 sends a TIMER signal, or a character is received on the optical serial line, the supervisor microcontroller IC8 switches the other parts of the device on (VKAP, VANA). Now the device is in **on** mode.

Then, in this **on** mode of the device, the supervisor microcontroller IC8 has mainly safety functions.

In its **on** mode, the ABPM-05 device is controlled by the main microcontroller IC1. The main microcontroller executes the program stored in the IC3 EPROM, and it uses the IC4 external RAM during runtime, too. It reads the current time from the IC6 real-time clock and it sets the real-time clock to send a TIMER signal after 1 minute. It communicates with the IC6 real-time clock and the IC7 LCD driver through the SCL and SDA of the I2C bus.

The measurement plan (monitoring schedule) and the results (blood pressure and pulse rate readings, event markers, service and error codes) are stored in the IC5 EEPROM. The main microcontroller can read and write the EEPROM using a simulated I2C bus, but writing has to be enabled on the DWP line handled by the supervisor microcontroller IC8, too. This can be accomplished by sending a command to the supervisor microcontroller.

The main microcontroller IC1 can communicate with the user's personal computer on an optical serial line. Communication is initiated by the user's personal computer in each case. The signals sent by the personal computer through one fiber of a twin plastic optical cable are received by the photodiode of the optical connector D3, they reach the receiver based

on the transistors T4 and T5, then the signal conditioner IC10A. The conditioned asynchronous serial signals are then received and interpreted by the main microcontroller. The asynchronous serial signals sent by the main microcontroller IC1 go through the signal conditioner IC10B to the LED of the optical connector D3, driven by the transistor T6. Optical signals sent by the device use the other fiber of the twin plastic optical cable to be transferred to the user's personal computer.

The pneumatic system of the ABPM-05 device consists of the motor pump MOT1, the operating valve V1, the safety valve V2, the pressure sensor IC17, the blood pressure measurement cuff, and the airway connecting components. Dust protection of the pneumatic system is ensured by dust separation filters, one on the suction side of the motor pump MOT1, another in the tube going to the cuff connector.

The piezo-resistive pressure sensor bridge IC17 converts pressure level in the pneumatic system to electric signal. The bridge is supplied by a current generator consisting of the active components IC11B, T7-T9. The output signal of the pressure sensor bridge is going through an amplifier-filter stage based on the operating amplifiers IC12, IC11A and IC11D to the internal AD converter of the main microcontroller IC1. The bridge balance at zero pressure level is set by the choice of appropriate value resistors for positions R32 and R33, while the amplification by the choice of an appropriate value resistor for position R41.

For the increased accuracy of pressure measurement, the oscillation signals are amplified the AC-coupled stage based on the IC11C operating amplifier, then they are converted by the internal AD converter of the main microcontroller IC1.

The supervisor microcontroller IC8 opens and closes the safety valve V2 due to the commands received from the main microcontroller IC1. For safety reasons, however, it keeps the safety valve closed for maximum approximately 3 minutes. If it does not receive a command from the main microcontroller to open the safety valve within this timeout, it opens the safety valve and resets the main microcontroller.

The operating valve V1 is controlled by the main microcontroller IC1. The valve will be opened if there are some smaller pressure strippings (low pressure) during the measurements.

The motor pump MOT1 is controlled by the main microcontroller as well. In order to be more silent, the motor usually operates on a lower voltage. In case of a higher pressure, the main microcontroller increases the voltage of the motor through a GEAR sign.

The buzzer BZ1 (which gives an acoustic acknowledgement if a pushbutton of the device is pressed) and the jumper JP1 (which serves only calibration purposes) are connected to the main microcontroller IC1.

At the appropriate command, the main microcontroller IC1 measures the voltage of the operating power source through the resistor network consisting of the resistors R76 and R75, and it writes the result to the LCD, thus making it possible for the user to check the remnant power of the AA size batteries.

The IC13 DC-DC transformer checks the inlet voltage through the R66, R67 resistance divider. If the values fall under the safe rate, the supervisor microcontroller IC8 switches the main microcontroller IC1 off, attaches the operation of the motor and the opens the closed valves. The main microcontroller will be kept switched off as long as the power supply is too low and it's operation will be only allowed if the power supply is back to the normal level.

ABPM-05 SERVICE INSTRUCTIONS**Tools:**

- Adjustable, current limited power supply (0.5-3V, 3A)
- Digital multimeter (min. 3 digits, 0.1-10V, 1mA-2A)
- Soldering iron, screw-driver, pincers
- Pressure meter (0-300 mmHg, ± 1 mmHg)
- Pump, valve and tubing
- PC and OI3 (or OI4) optical interface
- User program: CARDIOVISION v1.7 or later
- Service program: ABPM Service v1.0.0.26 or later

Chapters:

- SERVICE PROCEDURE
- CHECK PROCEDURES
- REPAIR METHODS
- TYPICAL PROBLEMS

SERVICE PROCEDURE

1. Record the problems

Ask the user to send or bring the faulty device to the service with its own accessories (at least the cuff). Faults of the accessories are frequent causes of problems. One of the most frequent faults is the leaking of the cuff.

The data and codes stored by the device when the problems occurred are necessary, too. It is best if the faulty device was not re-programmed and you can read data and codes directly from the device. Upload data from the device to your PC with the *user program* at once, so that they are not lost accidentally.

Make sure the problem indicated by the user really exists and you can reproduce it so that you can check whether it is eliminated as a result of repairs. If the fault is unknown, you have to do a device checkup to find the problem (E7.0).

2. Define the causes of the problems

Checks should always be started by checking basic functions. The instruction "Exchange device" in the **Check procedures** chapter (see later) means that this description does not include instructions for further troubleshooting and repairs. To decide whether another device should really be given to the user in exchange is not the objective of this description.

Displaying the service code on the LCD may be necessary for service purposes. The procedure to achieve this is purposefully difficult – you have to press and hold the extra measurement button while an error code is displayed on the LCD, and when the voltage appears on the LCD, you have to press the event button, too, then release both buttons.

3. Make repairs

If you have to open and disassemble the device, it is always worth cleaning the operating valve tightener.

4. Check the operation of the device

At the end of repairs, always make a full check of the device.

5. Document your work

All performed checks and repairs have to be documented in accordance with the usual practice of the given service. The report must include the serial number(s) of the device(s), the short description of the found problem(s), the performed check(s) and repair(s), a list of the used spare parts, the date of repairs and the signature of the responsible repairing person.

CHECK PROCEDURES**E1.0 Checking basic functions**

	What to do	Error	Elimination	Note
E1.1	Insert fully charged accumulators in the battery compartment	There is nothing on the LCD	Check the power supply (E2.0)	
		The voltage displayed on the LCD is lower than 2.4V	Use new batteries or fully charged accumulators	Using not fully recharged batteries may be misleading
E1.2	Wait for the disappearance of the voltage display			About 10 seconds
		The middle point in the lowest part of the LCD is displayed during the voltage display	Check the clock circuit (E4.0)	
E1.3	Press and release the event (heart) button	Nothing happens	Exchange device	The other button getting stuck may cause the problem, too
E1.4	Press and release the day-period (crescent moon) button	Nothing happens	Exchange device	The other button getting stuck may cause the problem, too
E1.5	Press and hold the measure (triangle) button	Nothing happens	Exchange device	
		Not all segments are displayed on the LCD	Exchange device	
		There are no beeps on pressing the button	Checking solderings on buzzer	
			Exchange device	

E2.0 Checking power supply

	What to do	Error	Elimination	Note
E2.1	Inspect the battery contacts	Deficiency, deformation, break, oxidation	Repair deformation or exchange battery contact(s)	
E2.2	Insert accumulators in the device and measure their voltage	<2.4V	Exchange the accumulators	
	Measure the voltage of the accumulators with a multimeter	Difference between the voltage measured by the multimeter and the device (as displayed on the LCD) is >0,2V	Exchange device	
E2.3	Measure the power consumption of the device	>800mA	Exchange device	

E3.0 Checking communication

	What to do	Error	Elimination	Note
E3.1	Start the <i>service program</i> , wait for connection	Connection does not work well or it does not work at all	Check settings and the interface	Follow the description of the <i>service program</i>
			Check the operation of the device (E1.0)	
			Exit other programs using the serial line, remove TSRs	
			Exchange device	

E4.0 Checking the clock circuit

	What to do	Error	Elimination	Note
E4.1	Insert accumulators in the battery compartment wait at least one hour, then set the clock of the device			Only if there have been no batteries in the device for months or after changing the button cell B1 supplying the clock IC
		Voltage display does not disappear and the crescent moon is displayed	Exchange the button cell B1 supplying the clock IC	Only in case of automatic voltage display after inserting the AA batteries
E4.2	Read the date and time of the device with the connection function of the <i>service program</i>	Connection fails	Check connection (E3.0)	The device stores the last two bits of the year only and the program calculates the year accordingly
		Wrong date or time	Adjust the clock of <u>the PC last used for programming the device or setting its clock</u>	
E4.3	Adjust the clock of your PC, then write date and time to the device with the time setting function of the <i>service program</i> . Read back the date and time you just wrote	Wrong date or time	Exchange device	The following points have to be carried out <u>only if</u> the problem is not likely to have been caused by the different settings of PCs
E4.4	Remove the accumulators. Re-insert them one day later and read the date and time of the device	Wrong date or time	Further check (E4.5). If it has been done already, exchange device	If there is no error, the E4.5 point does not have to be carried out.
E4.5	Remove the accumulators and measure the voltage of the internal button cell. Repeat this measurement one day later	<1.1V	Exchange the button cell and repeat the whole clock circuit check (E4.0)	You do not have to necessarily wait a full day. It is recommended to make a measurement from time to time because the problem usually turns out earlier.
			Exchange device	

The date or time is wrong if it differs from the real date and time by at least 3 minutes during one day. PC clocks are usually rather inaccurate, too!

E5.0 Checking the pneumatic system / airtightness

This stage checks the device and/or the cuff. If you want to check the device, apply a flawless, normal size cuff around an approximately 10-15 cm diameter cylinder with a rigid wall (artificial arm). When there is no pressure, there should be approximately half an inch free room between the cuff and the cylinder. A better solution is to use a high thermocapacity, ½ liter air tank instead of the cuff.

	What to do	Error	Elimination	Note
E5.1	Start the connection function of the <i>service program</i>	It does not work or it sends an error message	Check communication (E4.0)	
E5.2	Start the airtightness test of the <i>service program</i>	The device does not start pumping in 10 seconds.	Check communication (E4.0)	
			Exchange device	
E5.3	Wait. The device is pumping	Pumping does not end in 1 minute	Eliminate air leakage	*Cuff
E5.4	Wait. 2 seconds after pumping, the pressure should decrease to about 250 mmHg	>270 mmHg	Check and clean operating valve tightener	
			Exchange device	
		<220 mmHg	Eliminate air leakage	*Cuff
			Exchange device	
E5.5	Wait 1 minute. Leakage test	>3 mmHg with air tank, >6 mmHg if a cuff is used	Eliminate air leakage	*Cuff
E5.6	Wait. Deflation speed test	>1600 ms with air tank, >2400 ms if a cuff is used	Check and clean operating valve tightener	Check both values displayed by the <i>service program</i>
	Safety valve test.	>5 sec with air tank, >8 sec if a cuff is used	Check safety valve.	
E5.7	Start a new airtightness test but this time with a free air connector	The pressure displayed by the device is higher than 2 mmHg	Clean or exchange the filter in the tube to the cuff connector	
			Exchange device	

*Cuff: loss of air may be caused by a wrong cuff, too.

E6.0 Checking pressure measurement

Apply a flawless, normal size cuff around an approximately 10-15 cm diameter cylinder with a rigid wall (artificial arm). When there is no pressure, there should be approximately half an inch free room between the cuff and the cylinder. A better solution is to use a high thermocapacity, ½ liter air tank instead of the cuff. Connect the calibrated pressure meter, an air pump to increase pressure, and a valve to decrease pressure.

	What to do	Error	Elimination	Note
E6.1	*Decrease pressure to zero in the connected pneumatic system*			*This is very important! Always check with the calibrated pressure meter, too*
E6.2	Start the connection point of the <i>service program</i>	It does not work or sends an error message	Check communication (E4.0)	
E6.3	Start the pressure measuring mode point of the <i>service program</i>	The device does not show "v 0" and then " 0" on the LCD	Exchange device	Wait for 5-10 seconds to see this state
E6.4	Set 100 mmHg with the pump and the valve on your calibrated pressure meter	Difference between pressure meter and tested device is >3 mmHg	Exchange device	
E6.5	Set 200 mmHg	Difference is >3 mmHg	Exchange device	
E6.6	Set 300 mmHg	Difference is >3 mmHg	Exchange device	
E6.7	Exit pressure measuring mode	The pressure does not decrease, the device does not switch back to the normal mode	Check communication (E4.0) and the operation of the valves	
			Exchange device	

E7.0 Checking the device

	What to do	Error	Elimination	Note
E7.1	Check basic functions (E1.0)			
E7.2	Check power supply (E2.0).			
E7.3	Checking pneumatic system (E5.0)			
E7.4	Check pressure measurement (E6.0)			
E7.5	Start a blood pressure measurement and when pumping stops, disconnect the power supply of the device	Pressure does not decrease under 50 mmHg in 5 seconds	Exchange device	It is not necessary to measure accurately. Only the fast decrease of the pressure should be sensed
E7.6	Program the device with the <i>user program</i>	The device cannot be programmed	Check communication (E4.0)	The device should be programmed so that the first measurement be taken in some minutes from the moment of programming
		The device does not start measurements at the programmed time	See related point of the TYPICAL PROBLEMS chapter	
E7.7	Press the event button and measure your blood pressure. Record the result displayed on the LCD	The device does not measure	Exchange device	Do not move during the blood pressure measurement
	Upload and display the results with the <i>user program</i>	The measured result is not displayed or it is not the same as the one on the LCD	Exchange device	The problem may be caused by the fault of the user program, too, or possibly because the user program is damaged

REPAIR METHODS

J1.0 Airtightness problems

1. If the cuff is leaking, exchange the cuff.
2. If the device is leaking, check and point out the cause. If the system is leaking heavily, most frequently tubes are disconnected or loose. In case of minor leaking (5-20 mmHg in one minute), it is possible that one of the rubber components is worn or the motor pump is leaking. Separate small sections of the pneumatic system after pumping up to locate the leaking.

J2.0 Pushbuttons

1. If the buttons can be hardly operated or they get stuck, the reason is usually the contamination of the knobs. Disassemble the housing, remove the button knobs, clean the knobs, the buttons, and the housing around them.
2. The unreliable operation of the buttons can be the result of a wrong assembling of the device /housing/. Pay attention to careful assembling after repairs.

TYPICAL PROBLEMS

- **There is nothing on the LCD**

Check power supply (E2.0).

- **The device is on, but measurements cannot be started**

Check power supply (E2.0).

- **The device does not carry out the preset measurement program (schedule)**

Check the voltage of batteries in the device. Check power supply (E2.0). Check the functioning of buttons. Check the clock circuit (E4.0). Read out the results of the wrong measurements. If there are service codes, you can use them in searching for the error.

- **Complaint: „The device does measure but it does not store the results”**

It is often not an error of the device. It is a frequent complaint if the device starts the measurements at programmed times but it cannot complete them for some reason. This gives no or just a few valid readings (with BP values), but it may give the patient the impression that all measurements were carried out. The reason may be an error of the device, but it can be the low voltage of batteries or a leaking cuff, too. Checks and repairs should be based on the service codes you can read out from the device.

- **The device beeps a lot**

Read out the data of the device. If the device is programmed to measure at each minute or two minutes, the frequently attempted (and possibly unsuccessful) measurements may cause the frequent beeping, too. Checks and repairs should be based on the service codes you can read out from the device.

- **The device measures voltage inaccurately**

Measure the voltage of the batteries with a multimeter and compare them with the voltage measured by the device. A slight difference is acceptable (max. 0,2V) because the device does not measure exactly the same voltage.

- **The clock does not work**

Check power supply (E2.0). Check the clock circuit (E4.0).

- **Complaint: „The device cannot measure high blood pressure”**

Check the preset pressure limit with the user settings point of the *service program*. If it is too low, this caused the device to abort measurements. Check airtightness (E5.0). Measure with the cuff that belongs to the device because the problem is typically caused by the cuff.

ABPM-05 SERVICE INSTRUCTIONS FOR HOSPITAL SERVICES

The objective of this document is to guide technical personnel to check the operation of the Meditech ABPM-05 device, and to discern faults of the device and those of accessories. All checks described in this document are carried out without opening the housing of the device. Checks should be carried out with the accessories belonging to the device. However, you will need a reliable, perfect set of spare accessories to distinguish faults of the device and those of its accessories.

Avoid rescheduling the device if it contains data of an unsuccessful or otherwise suspect monitoring, and it may need to be repaired in an authorized Meditech service. These data may give valuable help to service personnel in locating faults.

Tools:

- PC and a set of ABPM-05 accessories
- Pressure meter (0-300 mmHg)
- Digital multimeter (min. 3 digits, 0.1-10V, 1mA-2A)
- Pump, valve and tubing
- User program: CARDIOVISION v1.7 or later
- Service program: ABPM Service v1.0.0.26 or later

The first column in the following tables contains checks to follow step by step. The second column describes possible errors. If you encounter one of these errors, the device is probably faulty. To decide if it really is or not, check the notes in the third column, because some conditions may imitate the error even with a perfect device.

1. Checking basic functions

	What to do	Error	Note
1.1	Insert 4 fully charged AA accumulators in the battery compartment of the device.	The LCD of the device remains blank or the displayed voltage is lower than 2.4V.	Accumulators may be faulty, too. Check them and repeat this check with another set of accumulators.
		Voltage display does not disappear within 15 seconds.	If there have been no accumulators in the device for weeks, the backup button cell inside may have run down. Wait an hour and recheck.
1.2	Check the voltage of the AA accumulators with a multimeter.	The difference between the voltage shown by the device and that measured with the multimeter is >0.3V	The consumption of the device slightly decreases the voltage of the accumulators as measured by the device.
1.3	Check the „stand-by” and the „pumping to open air” current consumption of the device. Then disconnect the multimeter from the device.	Current consumption >5mA in stand-by mode or >700mA while pumping to open air (no cuff connected).	The internal resistance of the multimeter disturbs the power supply of the device. If its internal resistance is too high, the device cannot even switch to stand-by mode.
1.4	Press the buttons and watch the LCD.	There is no beep on pressing a button or some LCD segments are faulty.	Disconnect the multimeter before this step because its internal resistance may disturb the operation of the device.
1.5	Connect the device to the PC, start the <i>service program</i> , choose Connect.	No connection.	Time should be on the LCD before the connection.
		The operation of the device is uncertain.	Another PC program (TSR) may disturb the connection.
1.6	Set the clock of the device with the clock-setting function of the <i>service program</i> . Read it back and recheck after 24 hours.	The date is wrong or the clock is more than 5 minutes late or fast.	If you do not suspect that the clock is faulty, it is enough to wait for a short time. Before setting the clock of the device, check and adjust the PC calendar and clock, too.
1.7	Program the device with the <i>User program</i> and wait for a measurement.	The device cannot be programmed or it does not measure at the programmed time.	Program the device only if you have not found any errors so far, because error codes in the device are deleted by the programming.
1.8	Press the event button, then the start button and measure your blood pressure	The device does not measure the blood pressure.	Apply cuff to your arm and do not move during the measurement. Record the result as seen on the LCD.
1.9	Read out and check the event and the reading with the <i>User program</i> .	The measurement or the event is missing or the result of the blood pressure measurement is wrong.	Data are not stored if the data memory of the device is already full.

2. Checking the pneumatic system, airtightness and measurement accuracy

This stage checks the device and/or the cuff. If you want to check the device only, apply a flawless, normal size cuff around an approximately 10-15 cm diameter cylinder with a rigid wall (artificial arm). When there is no pressure, there should be approximately half an inch free room between the cuff and the cylinder. A better solution is to use a high thermocapacity, ½ liter air tank instead of the cuff. Connect cuff or air tank to device. Connect device to PC.

	What to do	Error	Note
2.1	Start the airtightness test in the <i>service program</i> .	The device does not start pumping within 15 seconds.	Before the connection the time should be on the LCD.
2.2	Wait: the device is pumping.	Pumping does not stop in 1 minute	Leaking; may be caused by the fault of the cuff, too.
	Wait: 2 seconds after pumping, the pressure should decrease to about 250 mmHg.	Pressure is >270 mmHg or <220 mmHg	
	Wait 1 minute: leakage test.	>3 mmHg with air tank >6 mmHg with cuff around a cylinder	Leaking may also be caused by the fault of the cuff.
	Wait: deflation speed test.	>1600 ms with air tank >2400 ms with cuff around a cylinder	Check both displayed deflation values.
	Safety valve test.	>5 sec with air tank >8 sec with cuff around a cylinder	
2.3	Disconnect cuff/air tank. Start another airtightness test with free air connector.	The pressure displayed by the device is higher than 2 mmHg.	Nothing should block or obstruct the pneumatic system.
2.4	The pressure in the pneumatic system must be 0 before this step. Connect external pressure meter, pump and valve and check and ensure that pressure is 0. Then start the pressure measuring point of the <i>service program</i> .	The device does not switch to pressure measuring mode in 15 seconds.	
2.5	Set 100, 200 and 300 mmHg with the pump and the valve on the external pressure meter. Check LCD display on device.	Difference is >3 mmHg.	When defining the error, take the inaccuracy of your own pressure meter into consideration, too.

ABPM-05 service and error codes

The ABPM-05 device creates a 1 byte long service code in case of unsuccessful measurements or errors. This code refers to the cause of the error. Then the device creates a simplified error code from the service code. The device stores both codes in its EEPROM. The simplified error code will be displayed on the LCD, too. The user needs the simplified error code only. The service code is meant for service and development purposes. Any future modification of the internal program may affect the service codes without notice (e.g., new service codes may appear), but the error codes will be unchanged.

Service and error codes do not have an absolute certainty. Almost any code can occur due to electric disturbances, and movements of the patient may also imitate several kinds of errors, too. The need for absolute error identification is secondary to tolerability, patient compliance and safety. Therefore, the pneumatic system of the device may not be under pressure long enough to identify the cause of an error exactly. Codes confirming one another or the same codes occurring repeatedly are very likely to show the cause of the error properly.

Displaying the service code on the LCD may be necessary for service purposes. The procedure to achieve this is purposefully difficult – you have to press and hold the extra measurement button while an error code is displayed on the LCD, and when the voltage appears on the LCD, you have to press the event button, too, then release both buttons.

List of service codes *with corresponding error codes in italics* (simplified)

80	<i>90</i>	-	98	<i>1</i>	Operating valve error / patient motion
81	<i>9</i>	Supervisor microcontroller watchdog	99	<i>9</i>	-
82	<i>9</i>	Supervisor microcontroller SW watchdog	9A	<i>33</i>	Pumps too long (not for first time)
83	<i>9</i>	Main microcontroller watchdog	9B	<i>33</i>	Tries to pump to same level for fourth time
84	<i>99</i>	Pressure measured with open valves	...	<i>9</i>	-
85	<i>2</i>	Measurement interrupted manually	A1	<i>1</i>	Measurement timeout
86	<i>2</i>	Measurement interrupted manually	A3	<i>1</i>	Noisy measurement
87	<i>3</i>	Low voltage	A5	<i>34</i>	No oscillations
88	<i>31</i>	No pressure when pumping	A6	<i>1</i>	Invalid result
89	<i>32</i>	Clogged when pumping	A7	<i>1</i>	Too much motion
8A	<i>33</i>	Pumps too long when pumping	...	<i>9</i>	-
8B	<i>1</i>	Clogging (pumping)	B1	<i>2</i>	Button held for long time (measurement)
8C	<i>1</i>	Clogging (deflation)	B2	<i>2</i>	Button held for long time (event)
8D	<i>1</i>	Clogging (overpressure)	B3	<i>2</i>	Buttons held for long time (more)
8E	<i>1</i>	Too long measurement	B6	<i>2</i>	Button held for long time (day p.)
8F	<i>1</i>	Cannot set to pressure	...	<i>9</i>	-
90	<i>9</i>	-	C0	<i>90</i>	Supervisor microcontroller does not answer
91	<i>99</i>	High zero pressure	C1	<i>90</i>	Operating valve error
92	<i>8</i>	Pressure limit (motion)	C2	<i>90</i>	Safety valve error
93	<i>8</i>	Pressure limit (first time)	C3	<i>90</i>	EEPROM error
94	<i>8</i>	Pressure limit (second time)	C4	<i>90</i>	Wrong service block
95	<i>99</i>	No measurement because of prohibition	C5	<i>90</i>	RAM error
96	<i>4</i>	Supervisor microcontroller reset	C6	<i>90</i>	ROM error
97	<i>1</i>	Cannot set zero pressure level	C7	<i>90</i>	Calibration jumper connected
			...	<i>9</i>	-

List of error codes (simplified)

1	aborted measurement	31	cuff missing or loose
2	manually interrupted	32	cuff tubing clogged
3	battery rundown	33	cuff leaking or loose
4	batteries changed	34	cuff not on arm
8	pressure limit exceeded		
9	temporary disturbance	90, 99	device error

List of service codes with explanation *with corresponding error codes in italics*

The following list contains service codes and corresponding error codes. Numbered lines in the description part explain the actions that should be taken to identify and fix the causes of the given code. Actions to take are listed in the order of probability. First follow instruction #1, if this does not help, proceed to instruction #2, etc. Repairs should be done in authorized services only!

- 80 **90** UNUSED SERVICE CODE
1. No repairs necessary. If encountered several times, exchange device.
- 81 **9** The watchdog circuit of the supervisor microcontroller has signalled.
Operation of the supervisor microcontroller was disturbed by some electric signal but it resumed automatically.
1. No repairs necessary.
2. If encountered several times, check the soldering of the supervisor microcontroller. Exchange device if necessary.
- 82 **9** The watchdog program function of the supervisor microcontroller sensed an error.
Data of the supervisor microcontroller show that the main microcontroller has been in a given state for too long (10-150 sec, depending on state).
1. No repairs necessary.
2. If encountered several times, check the soldering of the supervisor microcontroller. Exchange device if necessary.
- 83 **9** The watchdog circuit of the main microcontroller has signalled.
Operation of the main microcontroller was disturbed by some electric signal but it resumed automatically.
1. No repairs necessary.
2. If encountered several times, check the soldering of the main and the supervisor microcontroller, the ROM and RAM. Exchange device if necessary.
- 84 **99** The device senses lasting pressure with open valves.
The value of the analog signal measured at zero pressure has to fall between given limits. Either the pressure measuring circuit went wrong or the device was not able to open the valves and it is under pressure. The latter case is highly improbable since the device has two valves controlled by two different controllers. The device checks the pressure continuously and if the above case persists, it does not start measurements. It tries to open the valves every minute, stores service and error codes, and displays error codes. The acoustic signal relating to the displayed error code cannot be disabled.
1. Exchange the device.

- 85 2 The measurement was interrupted by the patient.
 The patient pressed the measurement button during a measurement process.
 1. No repairs necessary.
 2. If it is not caused by the patient, check and repair the button.
- 86 2 The measurement was interrupted by the patient.
 The patient pressed the event or the day period button during a measurement process.
 1. No repairs necessary.
 2. If it is not caused by the patient, check and repair the button.
- 87 3 Low voltage.
 The measurement was aborted due to low voltage of the operating power source.
 1. No repairs necessary.
 2. Check voltage.
- 88 31 On the first pumping, the pressure is permanently too low.
 There is no cuff connected or the device is losing air heavily (e.g., a tube in the pneumatic system is disconnected).
 1. No repairs necessary.
 2. Check airtightness.
- 89 32 On the first pumping, the pressure increases extremely fast (clogging).
 If the air tank is too small, the pressure increases very fast on pumping (more than 100 mmHg in one second). Its reason is the clogging of the device or the cuff (e.g., there is some water in the device or the tube of the cuff is broken) or possibly – very rarely – the fault of the pressure measuring circuit.
 1. No repairs necessary.
 2. Check for clogging.
- 8A 33 On the first pumping, the motor pumps for too long.
 The device is losing air or there is a hole in the cuff.
 1. Change cuff
 2. Check airtightness.
- 8B 1 Clogging – by pumping.
 If the device strongly overpumps or overdeflates, it is likely that the tube of the cuff is broken. Since on the very first pumping the program directly checks for clogging (see service code 89), it is possible that this phenomenon is caused by the moving of the patient. Therefore, nothing happens on the first detection of this phenomenon, but two subsequent occurrences cause the device to abort the measurement and generate this service code. The reason of the error is probably the moving of the patient either directly or by breaking the tube of the cuff. This service code indicates that a measurement was aborted after an overpumping.
 1. No repairs necessary.
 2. Check for clogging.
- 8C 1 Clogging – by deflation.
 This is the same as service code 8B, except that the second phenomenon (when the measurement was aborted) was not an overpumping but an overdeflation. Contrary to service code 8B, this error can be caused by an error of the operating valve.
 1. No repairs necessary.
 2. Check for clogging.

- 8D **1** Clogging – overpressure.
 There is overpressure in the device but it is probably the side effect of the situation described at service code 8B. Probably, it was caused by the moving of the patient or the breaking of the cuff tube.
 1. No repairs necessary.
 2. Check for clogging.
- 8E **1** Measuring time safety check signal.
 The device checks measuring time at three levels (measuring algorithm, safety check of the main microcontroller, supervisor microcontroller), and this is the signal from the middle level. Since it is not the first (algorithm) level that signalled, the reason is some error, probably a temporary disturbance.
 1. No repairs necessary. If encountered several times, exchange device.
- 8F **1** Device regularly cannot step to required pressure level.
 The error is probably caused by the moving of the patient or less probably by some temporary clogging.
 1. No repairs necessary.
 2. Check for clogging.
- 90 **9** UNUSED SERVICE CODE
 1. No repairs necessary. If encountered several times, exchange device.
- 91 **99** The zero pressure is too high when starting the measurement.
 When starting the measurement, the value obtained on defining the zero pressure is too high. The pressure measuring circuit went wrong or the valves are closed and the device is under pressure.
 1. Exchange the device.
- 92 **8** Pressure over the preset limit for too long.
 The moving of the patient increased the pressure over the preset pressure limit for a prolonged period. The error is directly caused by the moving of the patient, but the original reason is the improper setting of the pressure limit. By increasing the preset limit, the error can be simply eliminated. The overpressure can possibly be caused by the clogging of the cuff tube, too.
 1. No repairs necessary.
- 93 **8** A pressure step over the pressure limit would be necessary for the measurement.
 The device tried to step to a pressure that the pressure limit does not allow.
 1. No repairs necessary.
- 94 **8** The device tried to pump above the pressure limit.
 When pumping to a pressure level, the pressure always decreases a little after the motor pump stops. To ensure the accurate setting, the device pumps a little over the required level. However, it cannot do so around the preset pressure limit because overpressure may cause the measurement to be aborted. Even in this case, it can usually reach the required pressure with maximum two attempts. This service code means that the device made several attempts but could not succeed. The service code may signal some minor leakage, too.
 1. No repairs necessary.
 2. Check airtightness.

- 95 99 The device did not measure due to internal prohibition.
 After two consequent error codes 99 (even if the service codes generating them were different), the device will be prohibited and it will not carry out automatic measurements. It does start a measurement if the measurement button is pressed, however. If it is able to measure blood pressure, prohibition will be cancelled. Rescheduling the device eliminates the prohibition, too.
 1. Exchange the device.
- 96 4 The operating power supply of the supervisor microcontroller failed.
 It indicates that the AA batteries were replaced. Another possible cause is a contact fault in the battery compartment.
 1. No repairs necessary.
 2. Check power supply related components.
- 97 1 Zero pressure level setting lasts too long.
 The patient was moving so much for a longer period that it was impossible to set the zero pressure level. The determination of zero pressure level is not sensitive to motion, so the extent of moving which can generate this service code is so high that it is not worth starting a measurement in such a case.
 1. No repairs necessary.
- 98 1 The operating valve is faulty and/or the patient moves too much.
 The opening of the operating valve is regularly followed by insufficient pressure decrease. Possible causes: a gag of the operating valve or too much patient movement. Data analysis confirmed that the second cause has an absolute majority. (The excess patient motion may, however, be the result of a partial gag of the valve, too, because a lasting constant or very slowly decreasing pressure can be so inconvenient that the patient reacts with moving.) See also service code C1.
 1. No repairs necessary.
 2. Check airtightness.
- 99 9 UNUSED SERVICE CODE
 1. No repairs necessary. If encountered several times, exchange device.
- 9A 33 The motor pumps for too long (not at the very first pumping).
 The error is similar to the error described at service code 8A, the difference is that it did not occur on the very first pumping, only later. The device is losing air or there is a hole in the cuff.
 1. Change cuff
 2. Check airtightness
- 9B 33 The device tried to pump to the same pressure level four times.
 Pressure drops between the attempts to pump to the same pressure step are normal. Possible causes: the device is not airtight, the cuff is leaking or - rarely – excess patient movement.
 1. Change cuff
 2. Check airtightness.
- 9C ...
 ... A0 9 UNUSED SERVICE CODES
 1. No repairs necessary. If encountered several times, exchange device.
- A1 1 Measurement timeout.
 The device could not measure the blood pressure within the time limit for a measurement. The reason is usually the moving of the patient.
 1. No repairs necessary.

- A2 1 UNUSED SERVICE CODE
 1. No repairs necessary. If encountered several times, exchange device.
- A3 1 Noisy measurement.
 The measurement failed because of the moving of the patient.
 1. No repairs necessary.
- A4 33 UNUSED SERVICE CODE
 1. No repairs necessary. If encountered several times, exchange device.
- A5 34 No oscillations.
 The patient did not wear the cuff.
 1. No repairs necessary.
- A6 1 The result of the measurement is physiologically impossible.
 If the patient is moving too much, it may very rarely happen that the device can apparently measure the blood pressure, but the result is invalid, physiologically impossible.
 1. No repairs necessary.
- A7 1 Buffers are full.
 The buffers storing detected pulses (and the noise imitating them) are full. This error may occur only under extreme circumstances, since buffer sizes are about one order of magnitude larger than the data they have to store in a normal case. By the time they could still overflow, some other control function almost always senses that the measurement should not be continued.
 1. No repairs necessary.
- A8 ...
- ... B0 9 UNUSED SERVICE CODES
 1. No repairs necessary. If encountered several times, exchange device.
- B1 2 The measurement button is pressed and held for a prolonged period.
 To generate this service code, the measurement button has to be in a depressed state for at least half a minute when the device should start a preprogrammed measurement. The error is usually caused by the patient (e.g., the patient lay on the device during sleep), but it can be the fault of the button, too.
 1. No repairs necessary.
 2. Check and repair the button.
- B2 2 The event button is pressed and held for a prolonged period.
 Causes are similar as with service code B1 but in this case it is not the measurement but the event button that is held depressed.
 2. Check and repair the button.
- B3 2 Both buttons are pressed and held for a prolonged period.
 Causes are similar to the one with service codes B1 or B2, but in this case at least two buttons are constantly depressed/pushed down.
 1. No repairs necessary.
 2. Check and repair the button.
- B4 ...
- ... B5 9 UNUSED SERVICE CODES
 1. No repairs necessary. If encountered several times, exchange device.
- B6 2 The day period button is pressed and held for a prolonged period.
 Causes are similar as with service code B1 but in this case it is not the measurement but the day period button that is held depressed.
 1. No repairs necessary.
 2. Check and repair the button.

- ... BF 9 UNUSED SERVICE CODES
1. No repairs necessary. If encountered several times, exchange device.
- C0 90 The main microcontroller cannot communicate with the supervisor microcontroller.
The main microcontroller sends a command to the supervisor microcontroller but there is no answer. The main microcontroller keeps trying until it succeeds. The device gives a beep about each 30 seconds and this cannot be disabled. The device will be prohibited, that is, it will not start automatic measurements.
1. Exchange the device.
- C1 90 The operating valve is faulty or gagged.
The operating valve does not decrease the pressure. Patient movement is an unlikely cause since other data indicate that the patient stays calm. See service code 98, too.
1. Check valve.
- C2 90 The safety valve is faulty or gagged.
The valve was found faulty during the usual safety valve check at the end of the measurement.
1. Exchange the device.
- C3 90 The EEPROM does not answer.
The EEPROM used for storing the measurements does not answer.
1. Exchange the device.
- C4 90 The service block of the EEPROM is corrupted.
The data necessary for the proper operation of the device are damaged.
1. Exchange the device.
- C5 90 RAM test reported a problem.
The automatic POST after inserting AA batteries found the external RAM faulty.
1. Exchange the device.
- C6 90 ROM test reported a problem.
The automatic POST after inserting AA batteries found the EPROM faulty.
1. Exchange the device.
- C7 90 Calibration jumper is connected.
The special jumper used for the calibration of the device is connected. This state is for calibration purposes, the device cannot measure blood pressure in this state. The device displays a special symbol instead of displaying the clock (see *LCD pictures*).
1. Exchange the device.
- C8 ...
- ... FF 90 UNUSED SERVICE CODES
1. No repairs necessary. If encountered several times, exchange device.

- - NO MEASUREMENT RESULT AND NO SERVICE/ERROR CODE EITHER:
In most cases, this is not caused by an error of the device:
 - The clock of the PC used for programming the device is incorrect and therefore the clock of the device is incorrect, too. This needs special attention with date settings, as they are not displayed on the LCD of the device.
 - The AA batteries are weak, their contact fails or there are no batteries in the device at all (at the preset measurement time).
 - The device was busy with some exclusive task other than measuring blood pressure at the time of the preset measurement (e.g., measuring the voltage of batteries or communicating with the PC).
- The device is faulty, e.g., the button cell of the real-time clock circuit, the EEPROM storing the results, etc. are wrong.

List of error codes with explanationUnsuccessful measurements

- | | | |
|---|-------------------------|--|
| 1 | aborted measurement | the measurement timeout is over, the measurement had to be aborted (the patient was moving) |
| 2 | manually interrupted | the measurement was stopped by pressing a button (the display differs from others: "OFF" on the LCD) |
| 3 | battery rundown | the AA batteries exhausted during measurement |
| 4 | batteries changed | the AA batteries were replaced during the measurement |
| 8 | pressure limit exceeded | the pressure in the pneumatic system exceeded the preset pressure limit |
| 9 | temporary disturbance | external electric signal (e.g., static discharge) disturbed the operation of the device |

Cuff related errors

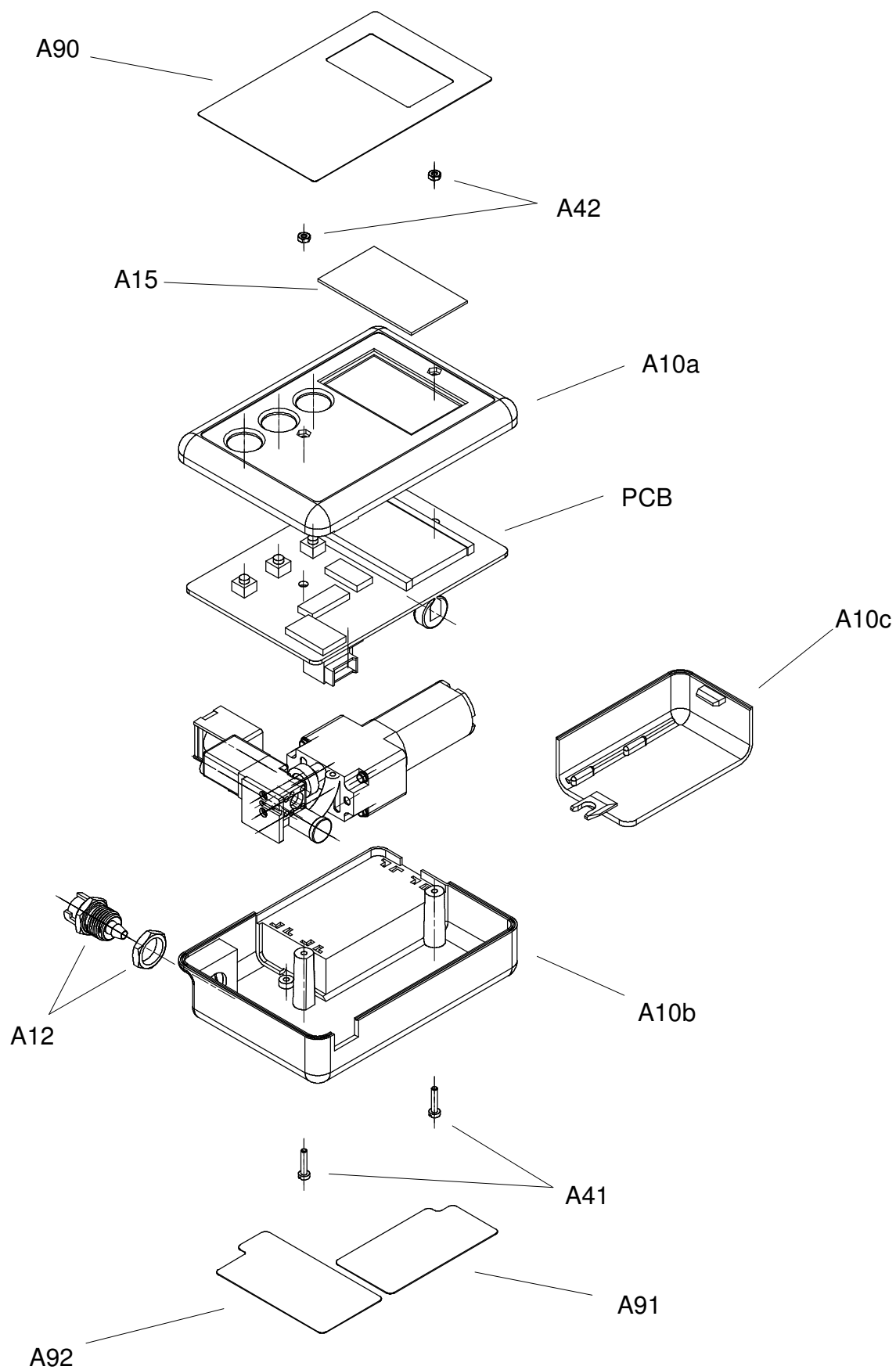
- | | | |
|----|-----------------------|---|
| 31 | cuff missing or loose | there was no cuff connected to the device, maybe the cuff is too loose on the patient's arm |
| 32 | cuff tubing clogged | the cuff is clogged or the rubber tube is broken |
| 33 | cuff leaking or loose | there is a hole in the cuff or it is very loose on the patient's arm |
| 34 | cuff not on arm | the patient did not wear the cuff |

Faulty device

- | | | |
|----|--------------|--|
| 90 | device error | the device could not measure due to a hardware error |
| 99 | device error | the device could not measure due to an error of pressure detection |

ABPM-05 service components

Order number	Identifier	Name
BP5I-R001	A30	Conductive rubber
BP5I-R002	LCD	LCD display
BP5I-R003	A15	LCD window
BP5I-R004	A10c	Battery cover
BP5I-R005	A41, A42	Screw set
BP5I-R006	B1	Accumulator
BP5I-R007	MOT1, C50, A31	Motor pump
BP5I-R008	V2, A32	Safety valve
BP5I-R010	A12	Air-connector
BP5I-R011	A22, A23	Motor filter
BP5I-R013	A11, A24	Special tube and filter
BP5I-R014	A50, A33	Valve module

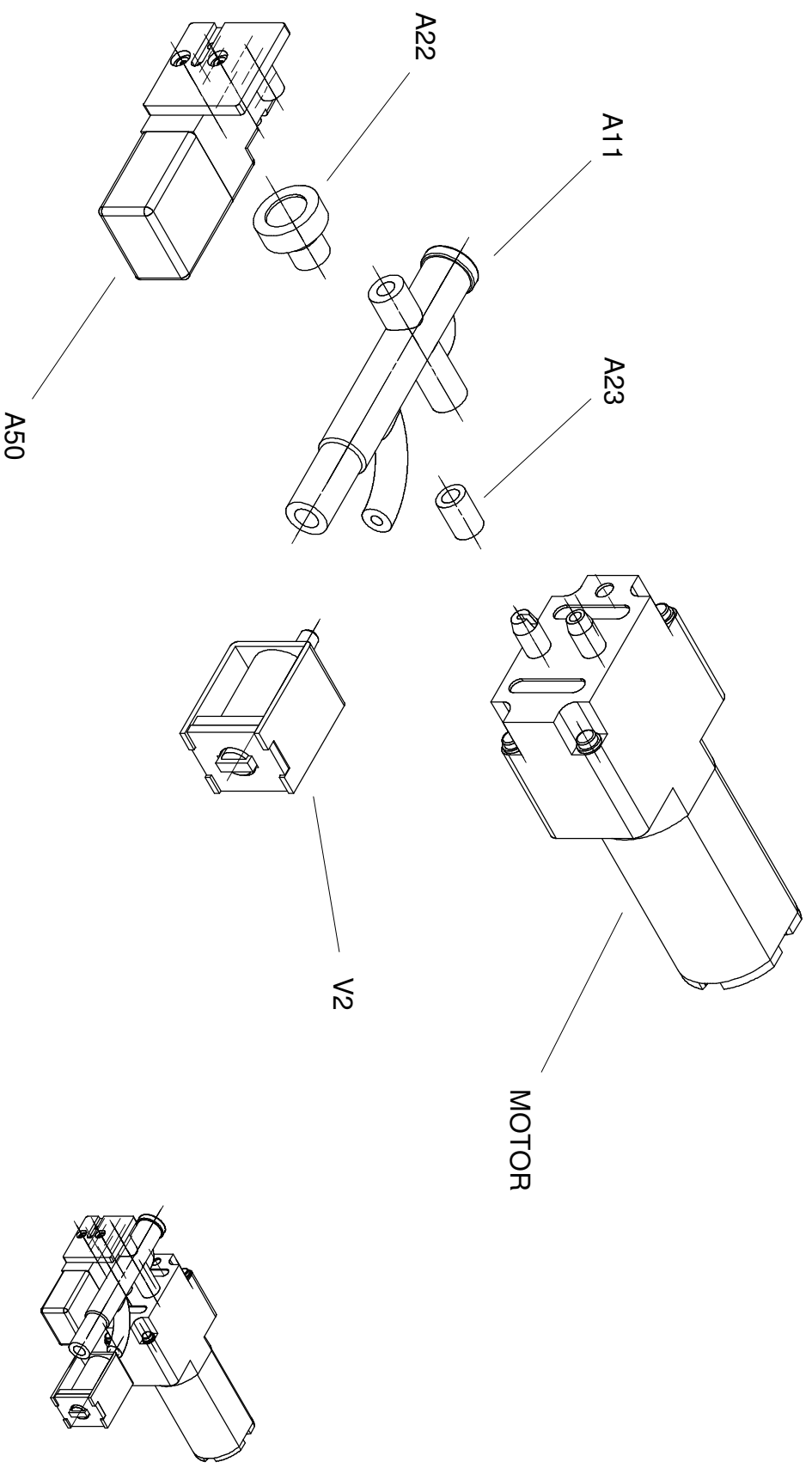


BP5N6601

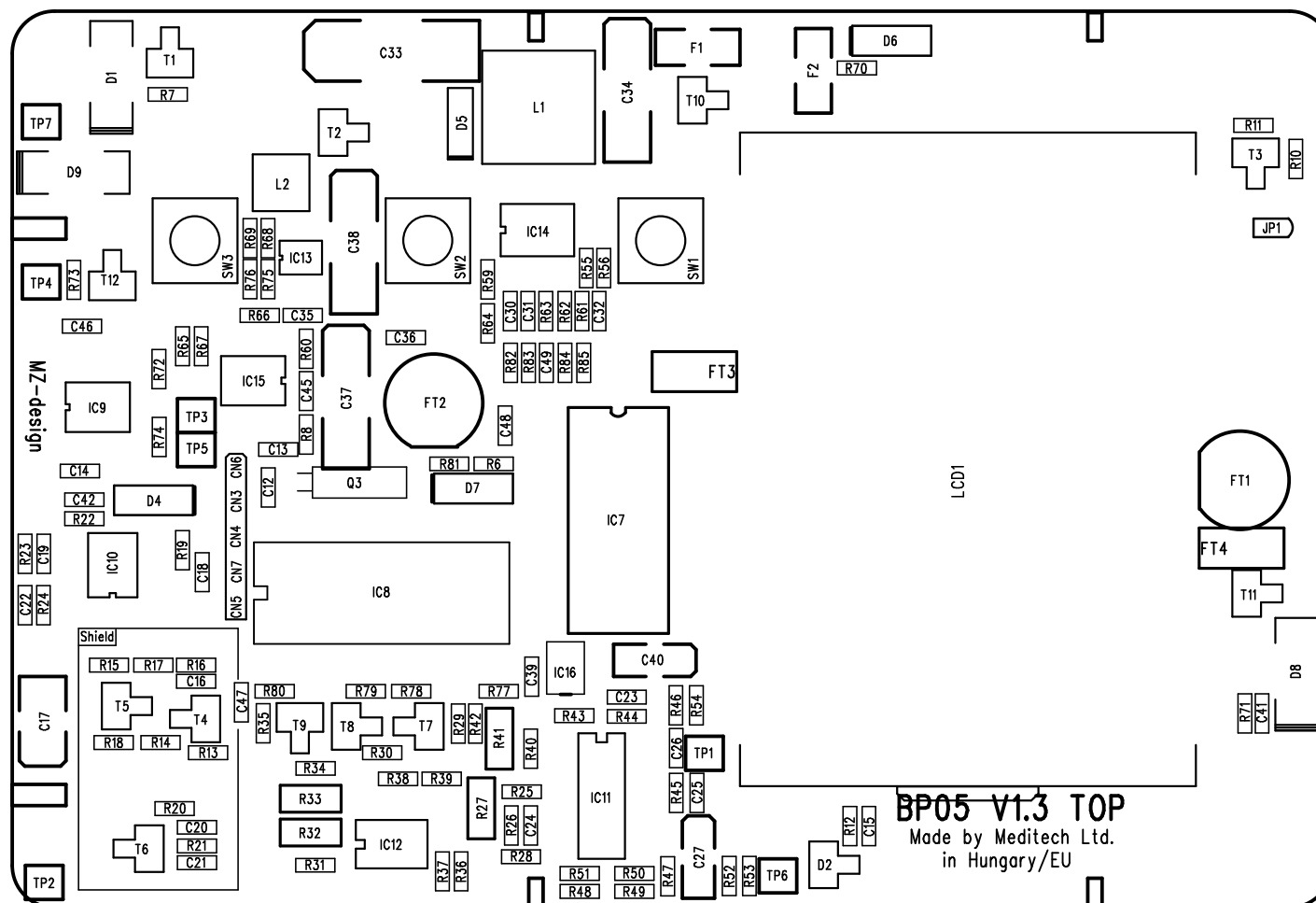
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BP5 exploded view

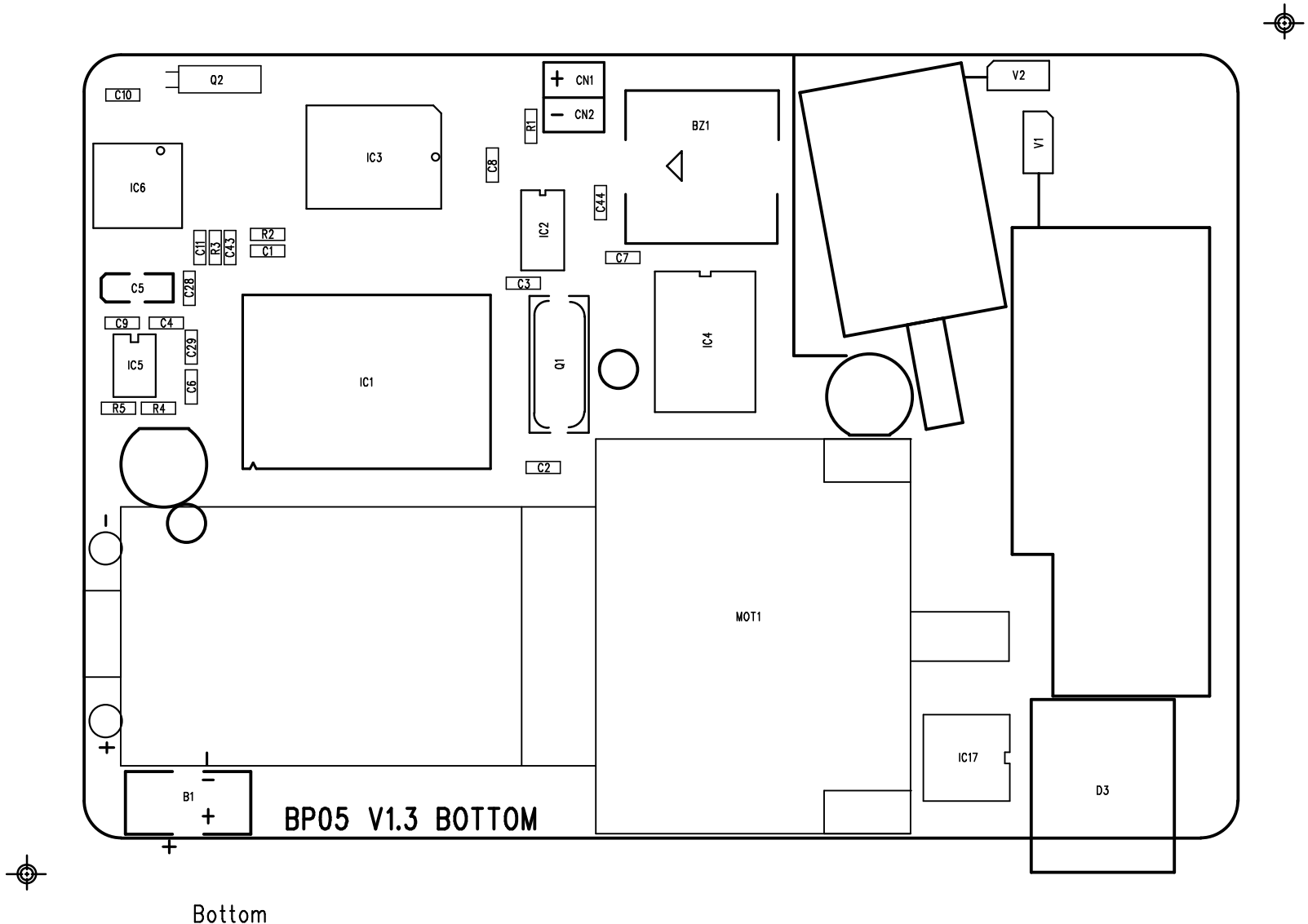
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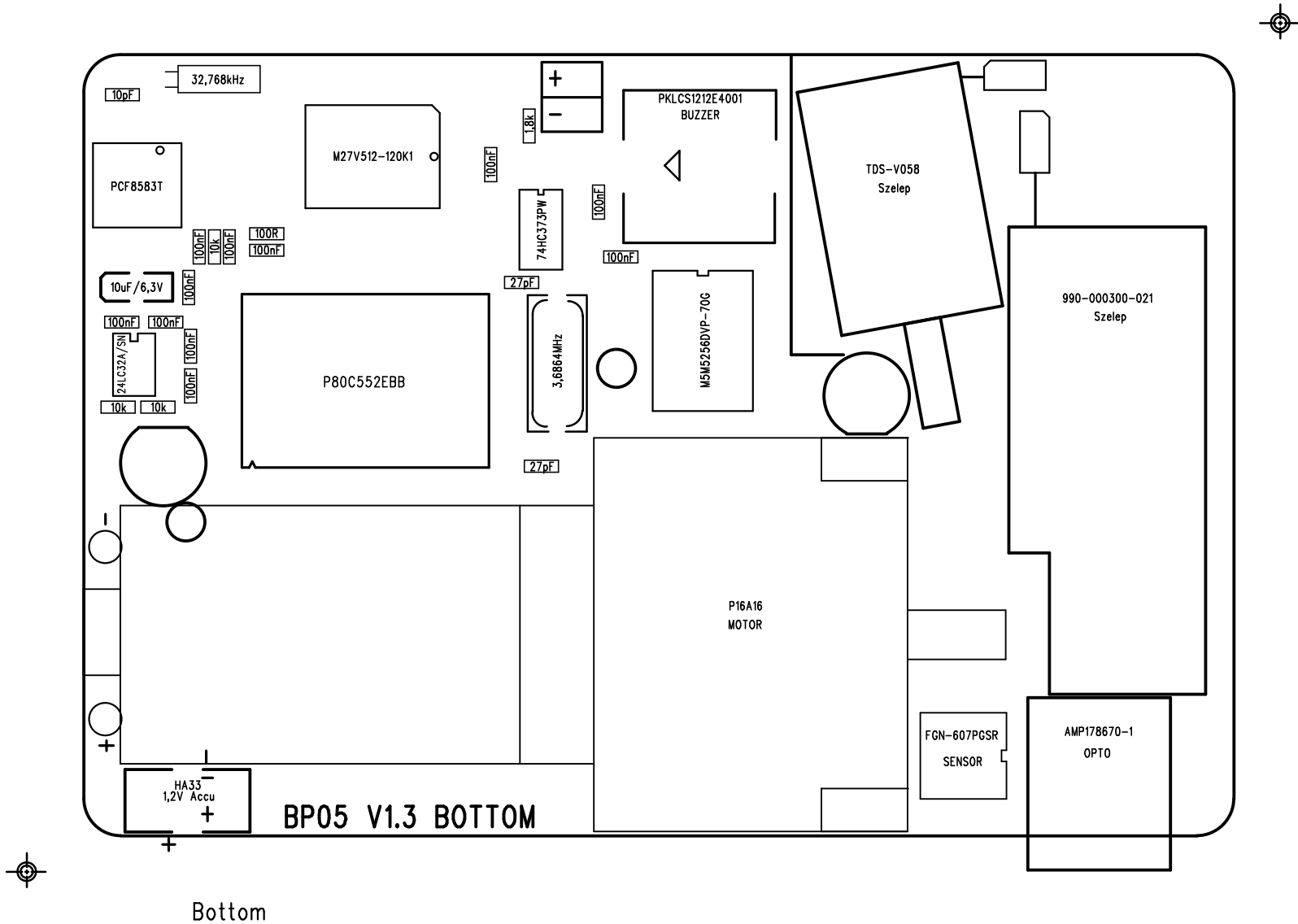


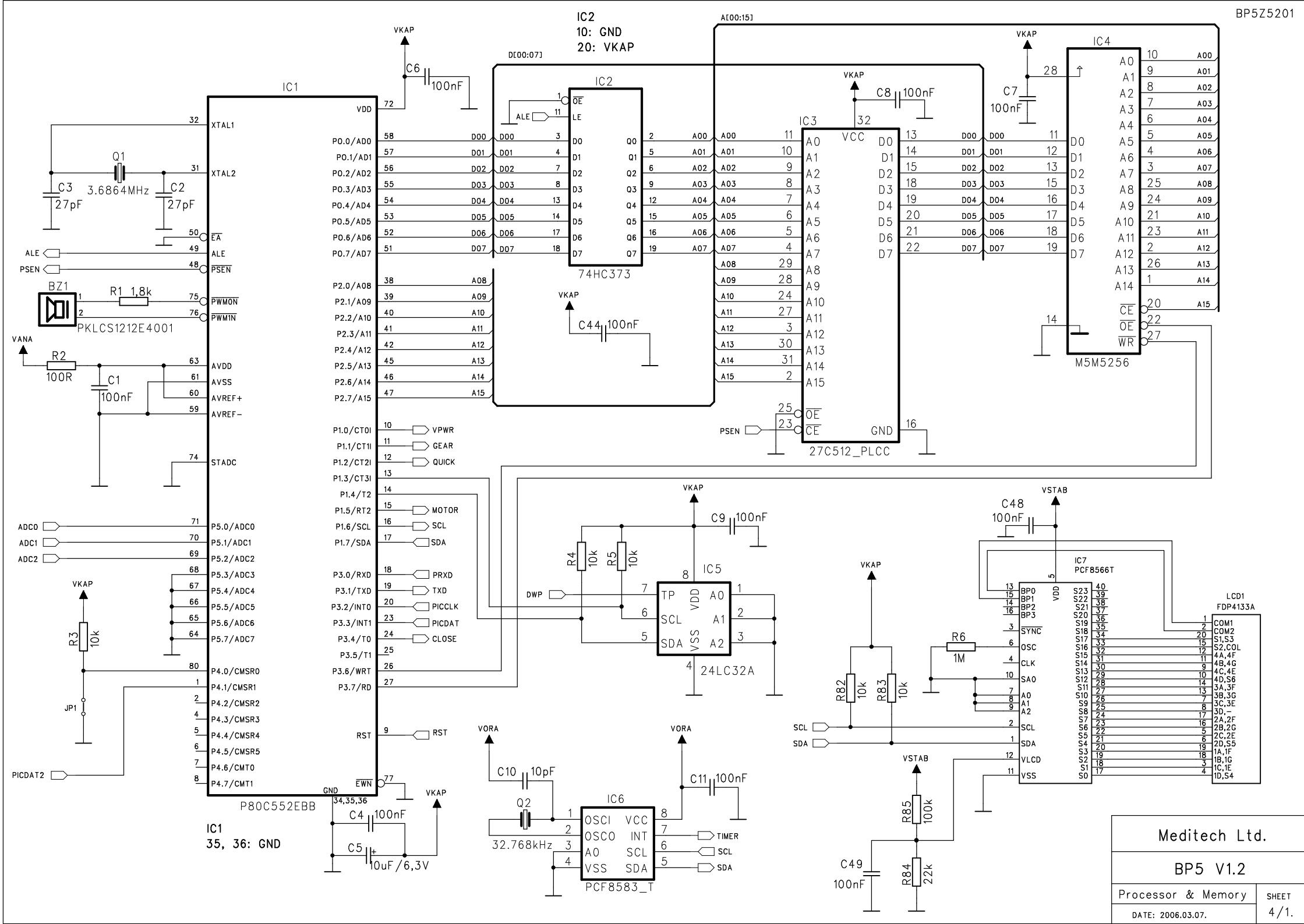
BP5N6601		BP5 exploded view	2/2
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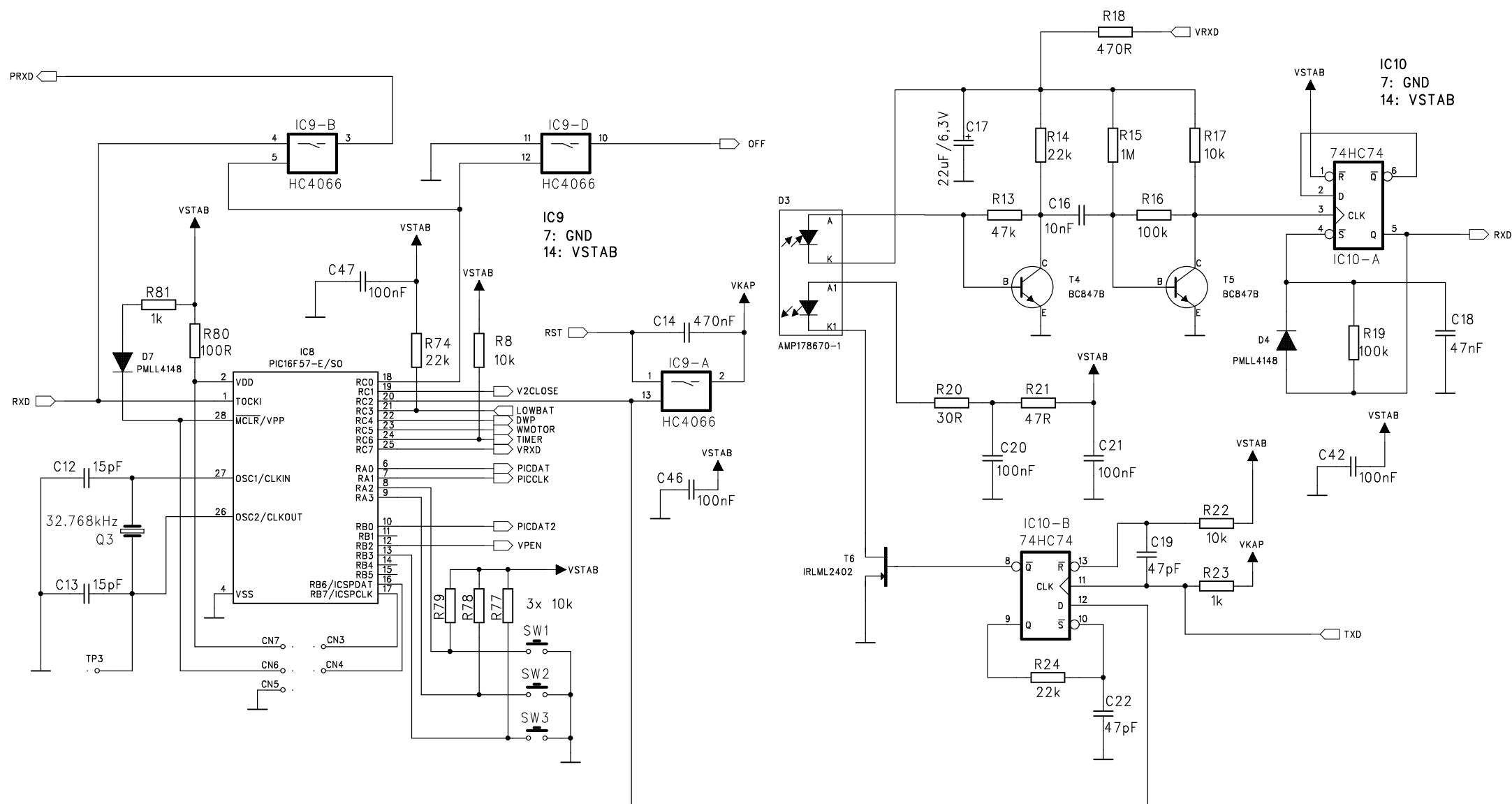
BP5 V1.2

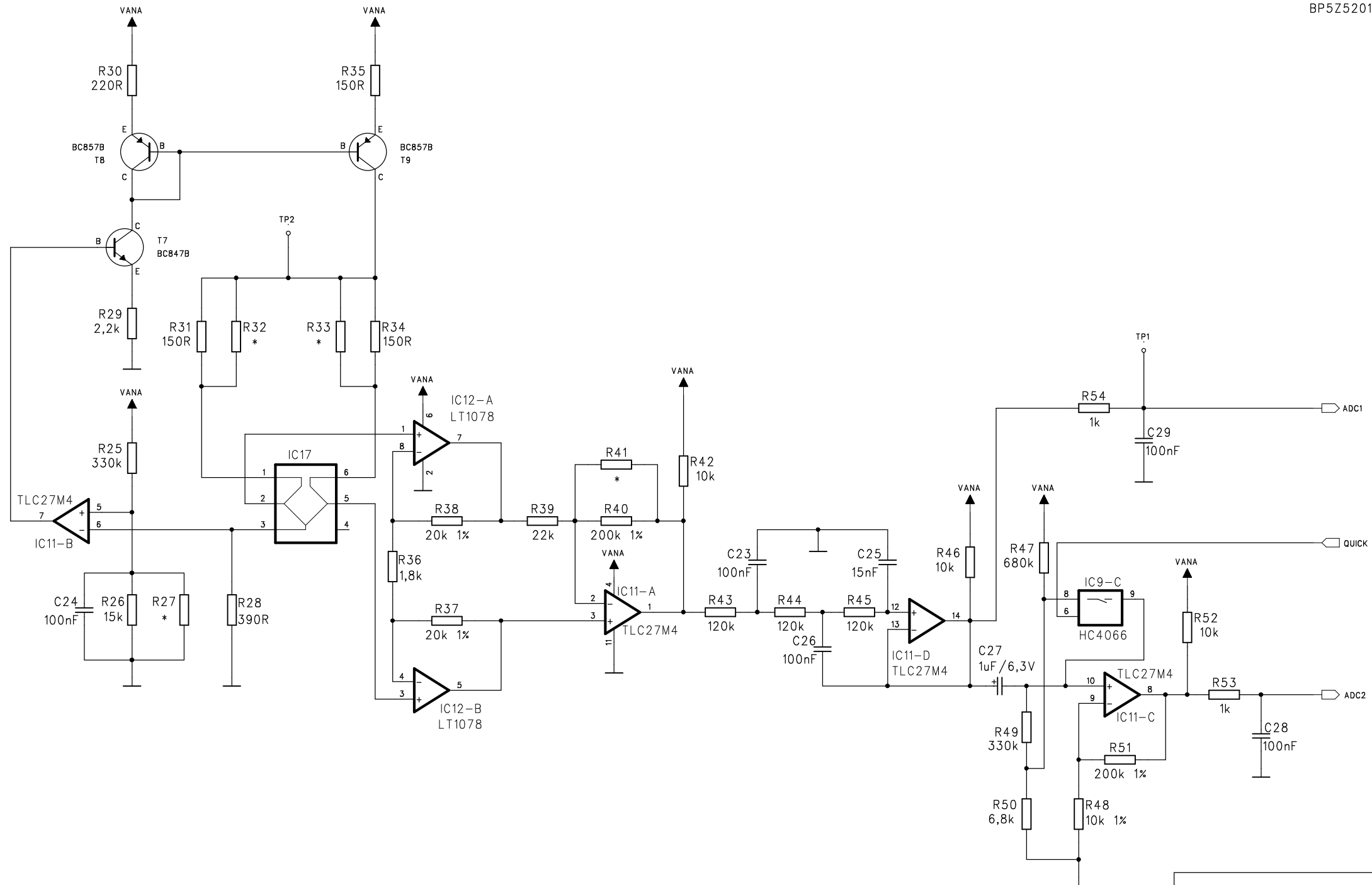
Processor & Memory

DATE: 2006.03.07.

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Functional diagram

